

Contact

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(LinkedIn)

Top Skills

Test Cases

Engine Control Unit (ECU)

System Testing

Languages

Maithili (Native or Bilingual)

Hindi (Full Professional)

English (Full Professional)

German (Elementary)

Certifications

E3. Series for Electrical Harness

Optimization algorithm using matlab

Self-Driving Cars Specialization

Study of Siemens PLC

Aerospace Engineering: Aircraft
Fundamentals and Advanced

Honors-Awards

Star of the Month

Team Award - Del-2 & Infantry.

DAV Student Scholarship

First Prize in Pitch Talk

First Prize in "Electrical Make-a-Thon
2k17"

Publications

BLDC Motor Driven Water Pump
Fed by Solar Photovoltaic Array
using Boost Converter

Border Detection System for
Fisherman using GPS and Arduino

Kumar Harsh

Robotics System Engineer | Nonlinear Control & System
Identification | evoBot | UAV Flight Controls | MIL/HIL Testing | Smart
Grids | Certification & Compliance (ANATEL, FAA, DO160)
Griesheim, Hesse, Germany

Summary

Systems Engineer specializing in UAV/UAS systems, aerospace electrical integration, verification & validation, and certification testing. Experienced in EMI/EMC compliance, HIL testing, battery qualification, system architecture, and wiring harness design for autonomous aerial platforms. Passionate about building reliable and scalable aerospace systems using robotics, control engineering, and embedded technologies.

Experience

WINGCOPTER

Ground Test Engineer

June 2024 - Present (2 years 1 month)

Weiterstadt

- Authored comprehensive Test Execution Plans (TEP), Qualification Test Procedures (QTP), verification reports, and formal compliance documentation for certification activities of battery, servos etc.
- Executed extensive battery characterization campaigns to determine critical voltage thresholds and State-of-Charge (SoC) mappings for Immediate Landing and Return-to-Home (RTH) safety logic.
- Designed and developed a simulated "dummy battery" architecture using modified BMS and relay PCBs with external resistor-divider circuitry, enabling destructive qualification testing without damaging production batteries.
- Led statistical analysis and development of UAV weight-envelope datasets using fleet-wide weight and balance reports to support repair conformity assessment activities.
- Performed EMI/EMC verification testing following software and hardware model updates to ensure continued system compliance.
- Led preparation of certification test plans, dry runs, conformity evidence, homologation documentation, and regulatory submission packages for ANATEL certification.

- Coordinated UAV conformity taskforce activities to align hardware configurations, BOM traceability, PCBA conformity, and certification evidence for homologation approval.
- Led EMI/EMC test campaigns and resolved compliance issues according to CISPR 32 and CISPR 22 standards.
- Conducted servo firmware Hardware-in-the-Loop (HIL) testing and verification data analysis aligned with DO-178 methodologies.
- Performed lighting system qualification testing in accordance with FAA, DO-160G, and ASTM standards.
- Tested SATCOM communication systems, analyzing latency, communication robustness, and UAV operational performance under different flight conditions to achieve Iridium Certification
- Analyzed multi-UAV operational datasets to establish production quality thresholds and lifetime test criteria for reliability monitoring.

Avenger Flight Group, LLC
 Flight Simulation Engineer
 May 2024 - June 2024 (2 months)
 Dreieich, Hesse, Germany

- Developed EASA-compliant Master-QTG documents for efficient maintenance of Airbus A330/A320 aircraft simulators

ie3 at TU Dortmund University
 Graduate Research Assistant (HiWi on Smart Grid Automation)
 December 2022 - April 2024 (1 year 5 months)

- Refined flexibility modeling of Intelligent Electronic Devices (IEDs) to adhere to IEC 61850 standards, optimizing for optimal control strategies
- Upgraded CIMPY 2.4 library to CIMPY 3.0 for IEC61970-301 standard
- Developed an automated framework for generating IID files using an ICD template for a given IED model in accordance with IEC 61850 standards
- Developed Python framework for streamlined creation of Digital Grid Connection Points and household auto-configuration (.scd) files, enhancing control efficiency in smart grids using CIM data.

Fraunhofer IML
 Master Thesis Student
 August 2023 - February 2024 (7 months)
 Dortmund, North Rhine-Westphalia, Germany

Topic: Nonlinear System Identification and Control using Sparse Identification of Nonlinear Dynamics for Model Predictive Control (SINDYc) with Evaluation on EvoBot.

Grade - 1.7

Master Thesis: <https://tu-dortmund.sciebo.de/s/qjEty9YHvae8zrC>

Codes: <https://github.com/hkharsh786/SINDYc-on-WIP.git>

Chair of Material Handling and Warehousing, TU Dortmund
Graduate Research Assistant (HiWi on Human Drone Interaction)
March 2022 - January 2024 (1 year 11 months)
Dortmund, North Rhine-Westphalia, Germany

- Contributed to creating a 6Dof dataset using IMU data for training the Drone Swarm's gesture control model
- Optimized location tracking accuracy through PID tuning and sensor fusion using EKF, integrating IR camera and IMU data enabling automated battery charging during swarming operations
- Contributed to the migration of a ROS based swarm show platform to ROS-2 using crazyswarm for indoor logistics application
- Developing program Nao and Robo-master robots for various showcases and 'Block-chain Europe' projects
- Enhanced pallet detection by fine-tuning Crazyflie drone controllers, ensuring stability for clear imaging with the integrated camera deck.
- Utilized the crazyswarm platform to support and troubleshoot drone shows for an indoor logistics application.

TATA Advanced Systems Limited

2 years 3 months

Executive- Aircraft Design

June 2020 - August 2021 (1 year 3 months)

Bengaluru, Karnataka, India

- Applied for the patent with company tie-up for "Eco-Friendly Cars, using Artificial Photosynthesis"
- Worked on designing a prototype of "Invisible UAV to Naked Eyes"

- Small UAVs:

- o Worked on overall product enhancement from the perspective of electrical LRU interfaces and improving product reliability.

- o Responsible for selection of BLDC motor, endurance estimation, loss calculation, and selection of required electrical wires and connectors for UAVs.
- o Been a member of a team and developed i-VTOL UAV, later bagging the "Best Team" Award.
- o Increased the accuracy of target tracking by introducing an Extended Kalman Filter to smoothen sensor values for camera-gimbal stabilization.
- o Responsible for creating BOM, System Architecture for LRU interface, breakpoint interface connection, wiring schematic, and routing design for all UAVs, Gimbals, and improvising products.

- Large Airborne System:

- o Developed autopilot control laws and system architecture for Medium Altitude Long Endurance (MALE) Aircraft. Also formulated Extended Kalman Filter (EKF) for altitude estimation
- o Designed and tuned PID controller to improve the stability characteristics of the aircraft
- o Testing of control strategy for robustness in MIL and SIL environment
- o Responsible for BOM creation and Man-hour Effort Estimation (Costing) of Sikorsky S-76 helicopters and Airbus C295 aircraft.
- o Responsible for Understanding and preparing the System Architecture design of SPn-P3 aircraft.
- o Designed EMI/EMC qualified Electrical Wiring Harness for MALE UAV.

Graduate Engineering Trainee

June 2019 - June 2020 (1 year 1 month)

Bengaluru, Karnataka, India

- Designed a precision-focused, laser-powered automated jig for missile wing calibration
- Worked on designing and tuning PID controllers to improve the stability characteristics of small UAVs meeting MIL-F-8785-C standards.
- Worked on UAV electrical system integration to improve product overall performance and reliability.
- Worked on feasibility analysis of Hydrogen fuel cell in UAV (16Kg Class) for better endurance

Tata Steel

Vocational Trainee

June 2018 - July 2018 (2 months)

Jamshedpur Area, India

PROJECT:

Speed Control of motor

Observed practical working of Motors in several plant in the industry.

Project Description:

Designed a vehicle having automated speed control. The model vehicle had an ability to control speed as per its distance from the obstacle ahead to avoid collision. Also the system used a digital feedback and used the control-ability of model using digitally controlled voltage supply to the motor (wheels).

National Institute of Technology Jamshedpur

Research Intern

May 2018 - June 2018 (2 months)

Jamshedpur Area, India

Worked on BLDC and designed an automatic solar power based Water Pump. Modeled and simulated Automatic Solar Water Pump using Moisture Level feedback of soil.

Internshala

Campus Ambassador

August 2017 - March 2018 (8 months)

India

Connected around 1000 Students to Internshala Platform.

Organized two days Internship Fest , where approx 400 students got internships.

Tata Steel

Vocational Trainee

June 2017 - July 2017 (2 months)

Jamshedpur Area, India

Study of Siemens PLC

- ✓ Had some Hands on Experience on use of PLC in Industries
- ✓ Design of program for automatic mixing of components for blast furnace fuel mixture (coal, coke etc) and monitoring its temperature rise using PLC S7-300 and S7-400.
- ✓ Designed and E-learning module on Siemens PLC S7-300 and S7-400

Rohan Motors Ltd

Marketing Intern

June 2016 - June 2016 (1 month)

Noida Area, India

Worked on marketing and sales policy.

Jai Yogeshwar Charitable Trust

Ngo

December 2015 - December 2015 (1 month)

Bihar

Published Swachh Bharat Abhiyan and made peoples aware in five villages.

Education

University of Dortmund

Master of Science - MS, Automation and Robotics · (October 2021 - February 2024)

Vellore Institute of Technology

Bachelor of Technology, Electrical and Electronics Engineering · (July 2015 - May 2019)

DAV PUBLIC SCHOOL, BISTUPUR

Intermediate, Science · (2001 - 2014)